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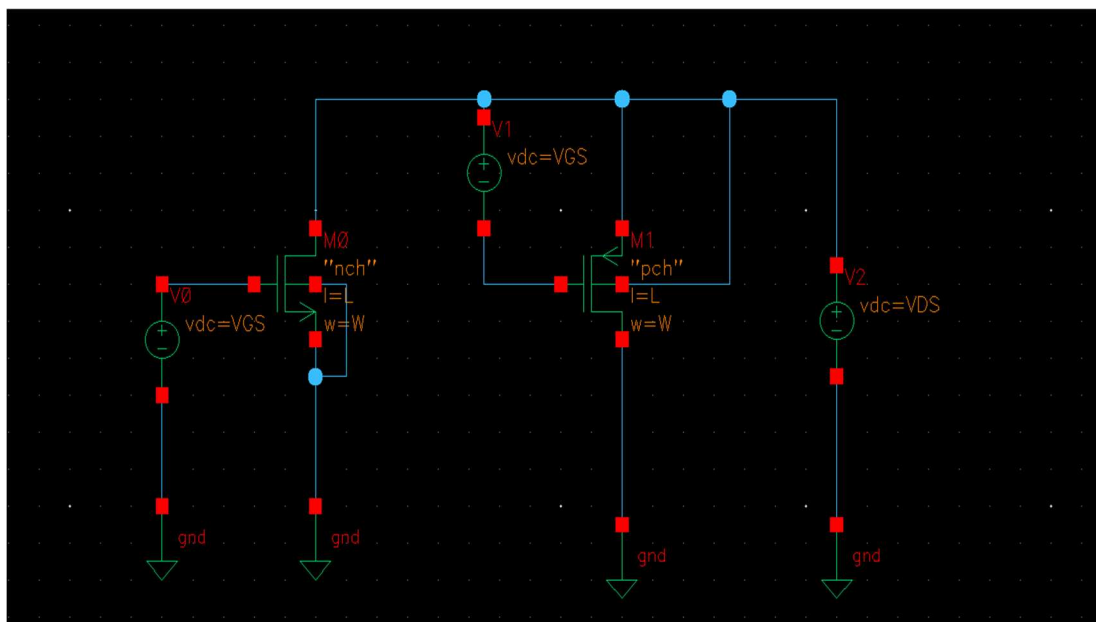
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Task 2

Part 1: Sizing Chart

DC Operating point Analysis

Schematic Test Bench



Given parameters

DC gain = -9

VDD = 1.8V

Biasing current = 84 μ A

Length = 2 μ m

Calculator.net

home / math / random number generator

Random Number Generator

This version of the generator creates a random integer. It can thousand digits.

Result

84

Lower Limit

Upper Limit

Biasing current

Calculator.net

FINANCIAL

home / math / random number generator

Random Number Generator

This version of the generator creates a random integer. It can deal with very large into thousand digits.

Result

9

Lower Limit

Upper Limit

DC gain

$$A_v = -g_m R_D = -\frac{2I_D R_D}{V_{OV}} = -\frac{2V_{RD}}{V_{OV}}$$

Let $g_m = \frac{2I_D}{V^*}$, where $V^* \approx V_{OV}$

$$g_m = \frac{2V_{RD}}{V^*}$$

$$V_{RD} = \frac{V_{DD}}{2} = \frac{1.8}{2} = 0.9 \text{ V}$$

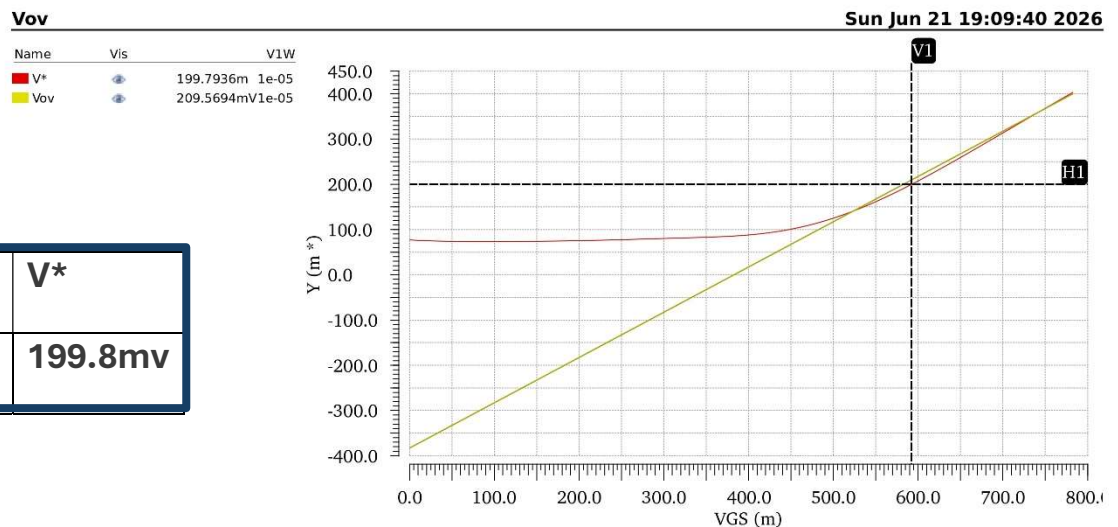
$$9 = \frac{2 \times 0.9}{V^*}$$

$$V^* = \frac{1.8}{9} = 0.2 \text{ V}$$

$$\therefore V^* = 200 \text{ mV}$$

We know that the current depend on W and gm. Rd depends on voltages of Vrd and Vov so the Av is independent of W so we can assume any W to get the Vov q and Vgs q and then change the W so we get the right biasing current .

V* and Vov vs VGS

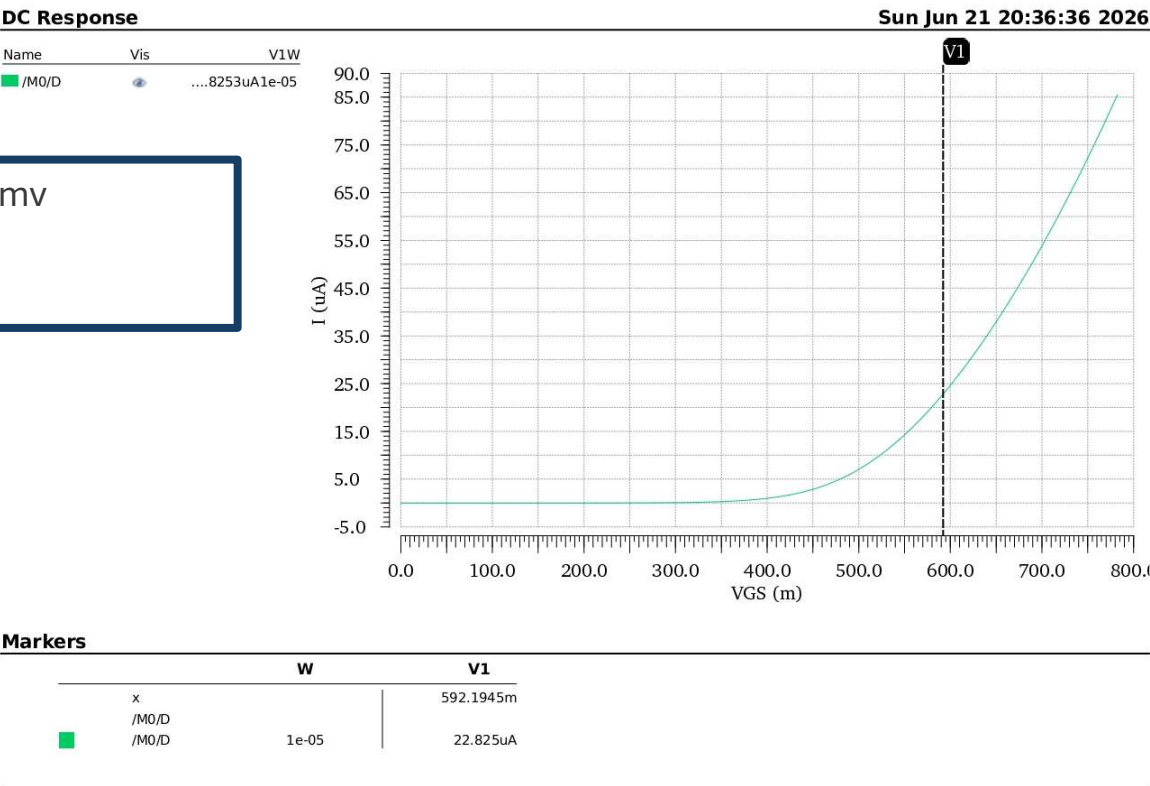


Vgsq	Vovq	V*
592.19mv	209.56mv	199.8mv

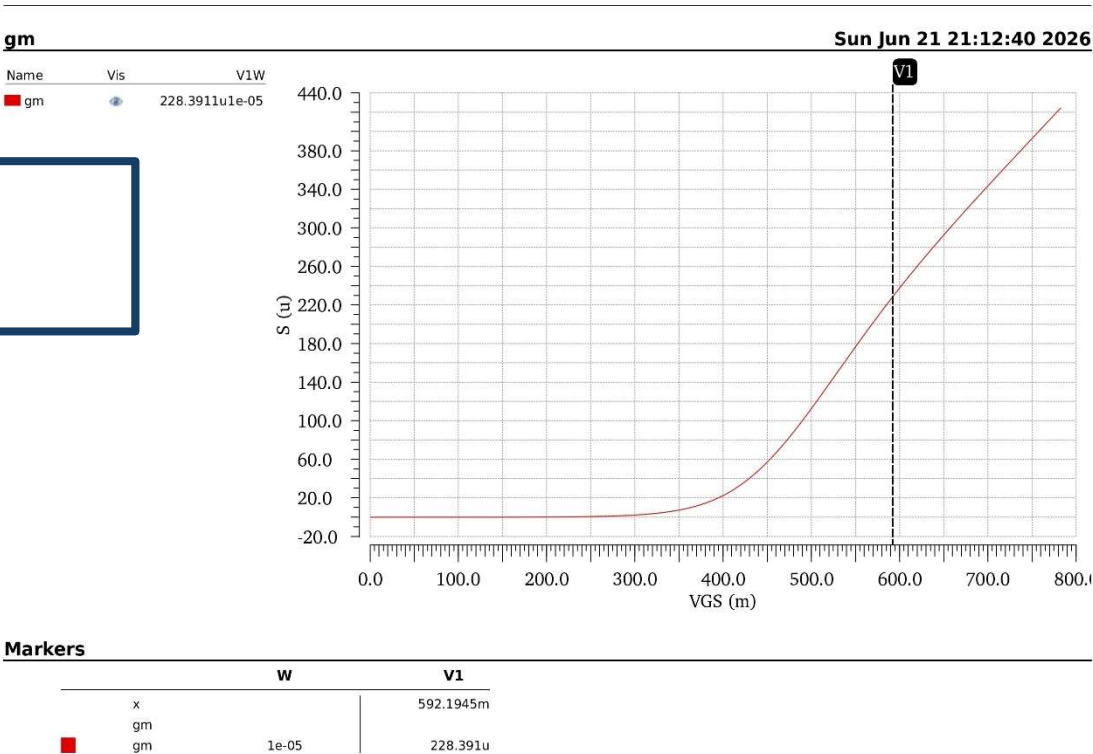
Markers		W	V1
	x		592.1945m
	V*		
	V*	1e-05	199.7936m
	Vov		
	Vov	1e-05	209.5694mV

MOSFET Operating Point at Width=10um

IDx vs VGS at w= 10um



gm_x vs VGS at w=10um



gdsx vs VGS at

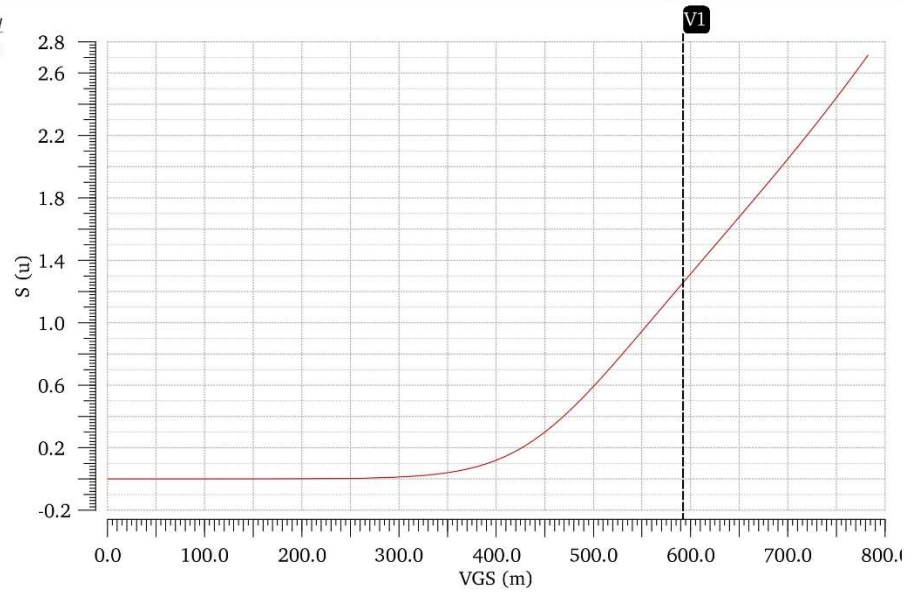
gds

Sun Jun 21 21:12:40 2026

Name	Vis	V1W
gds		1.25633u1e-05

VGS= 592.1945 mv

gds=1.256uS



Markers

	W	V1
x		592.1945m
gds		
gds	1e-05	1.25633u

	At x point	At operating point
ID	22.825uA	84uA
W	10um	36.80175um

	At x point	At operating point
gm	228.3391uS	840.327uS
W	10um	36.80175um

	At x point	At operating point
gds	1.256uS	4.63uS
W	10um	36.80175um

$$R_D = \frac{V_{RD}}{I_D} = 10.714 \text{ k}\Omega, \quad r_o = \frac{1}{g_{as}} = 215.58 \text{ k}\Omega, \quad R_D \parallel r_o = 10.207 \text{ k}$$

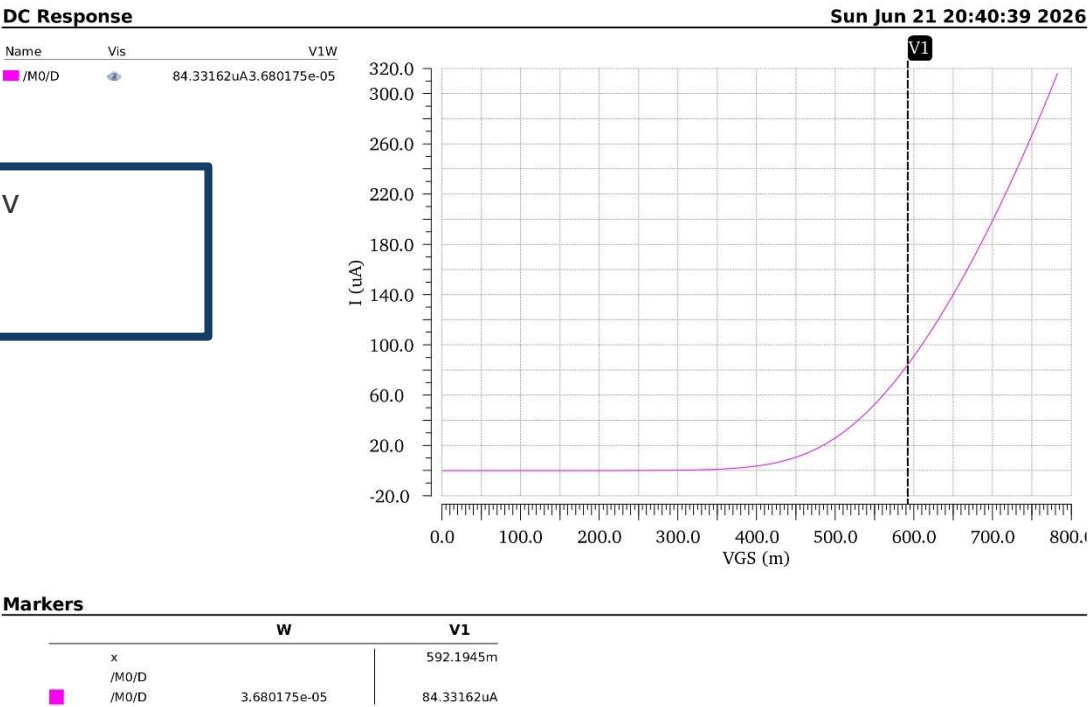
$$A_v = -g_m(R_D \parallel r_o) = -(843.915 \mu\text{S})(10.207 \text{ k}\Omega)$$

$$A_v = -8.61 \text{ V}$$

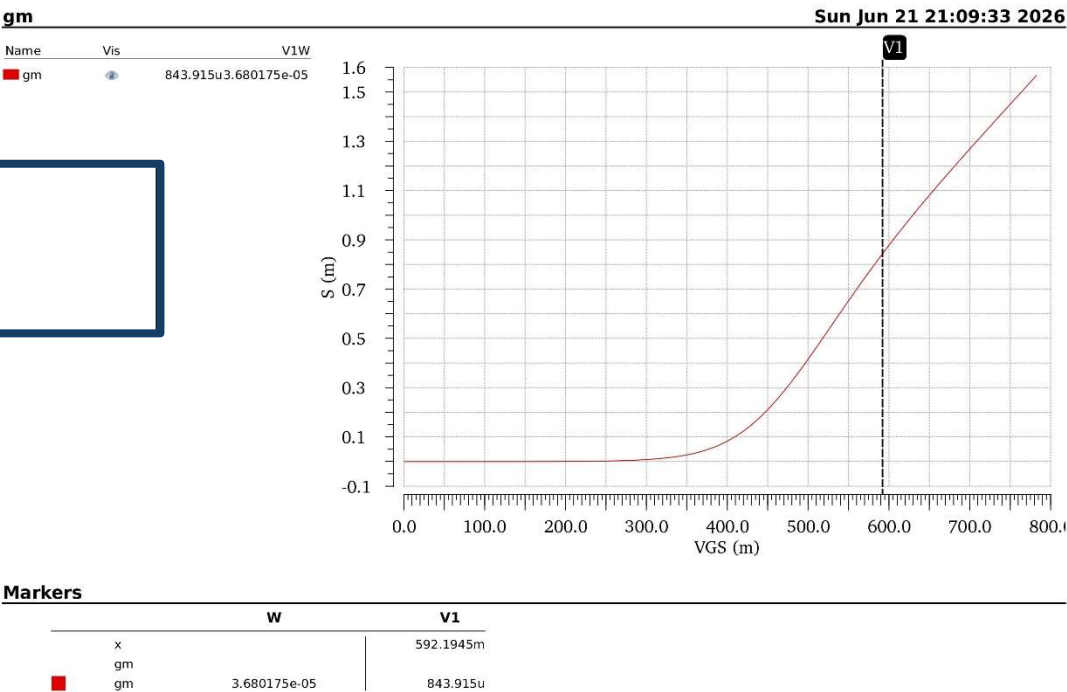
Note: The difference is
due to neglecting Rout

MOSFET Operating Point at Width=36.80175um (Biasing point)

IDq vs VGS at w= 36.80175um



gmq vs VGS at w= 36.80175um



gdsq vs VGS at w= 36.80175um

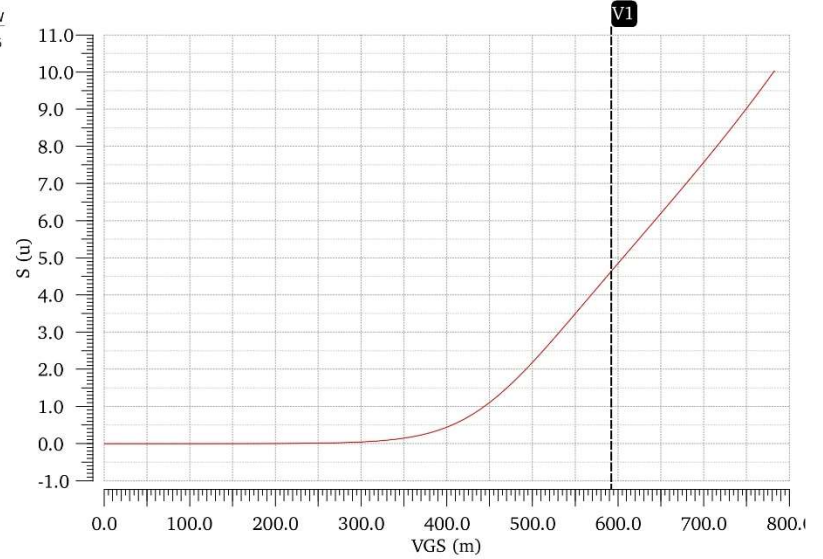
gds

Sun Jun 21 21:09:33 2026

Name	Vis	V1W
gds		4.63867u3.680175e-05

VGS= 592.1945 mv

$g_{ds}=4.63\mu S$



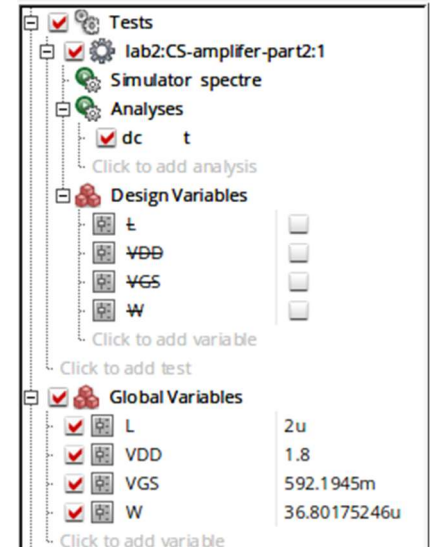
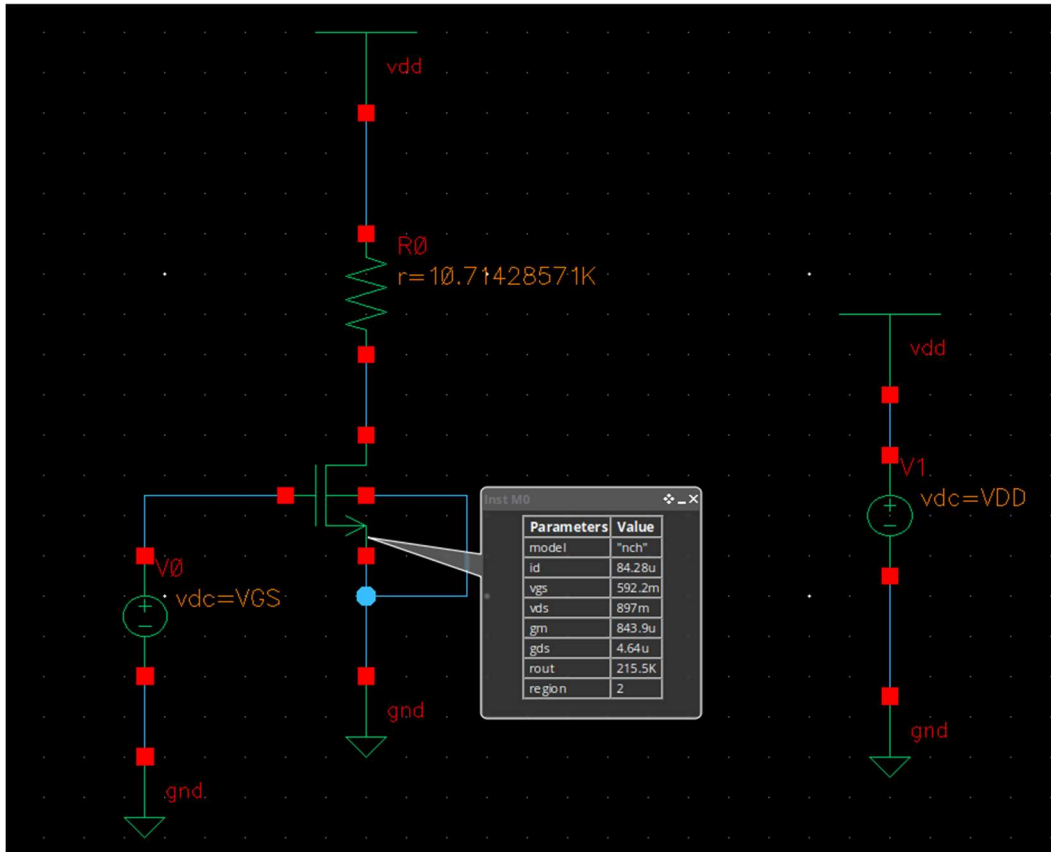
Markers

	W	V1
x		592.1945m
gds		
gds	3.680175e-05	4.63867u

Part 2: CS Amplifier

Operating Point & AC Analysis

Schematic Test Bench



Sizing chart values

ID=84.3uA

VGSq=592.1945 mv

VDSq=0.9

gmq=843.915us

gdsq=4.63uS

Note: The sizing chart values are very close to the ballon parameters, region 2 is saturation

3) Comparison between Rout and Rd

r_o	R_d
215.5kohm	10.714kohm
parasitic	Design choice
Can be ignored	$r_o R_d \approx R_d$

Ignoring r_o is justified in this case as $r_o \gg R_d$ so $r_o || R_d \approx R_d$ so $A_v = g_m R_d$ if the L was smaller the approx. may not be valid as $L \propto V_a \propto r_o$ so if L decreased the r_o will decrease as a result of CLM effect becomes stronger so the approx. may not be valid and the error will increase

4) Intrinsic gain

$$\text{intrinsic gain} = g_m \cdot r_o = 843.915 \mu\text{S} \cdot 215.5 \text{ k}\Omega = 181.86$$

5) Amplifier gain

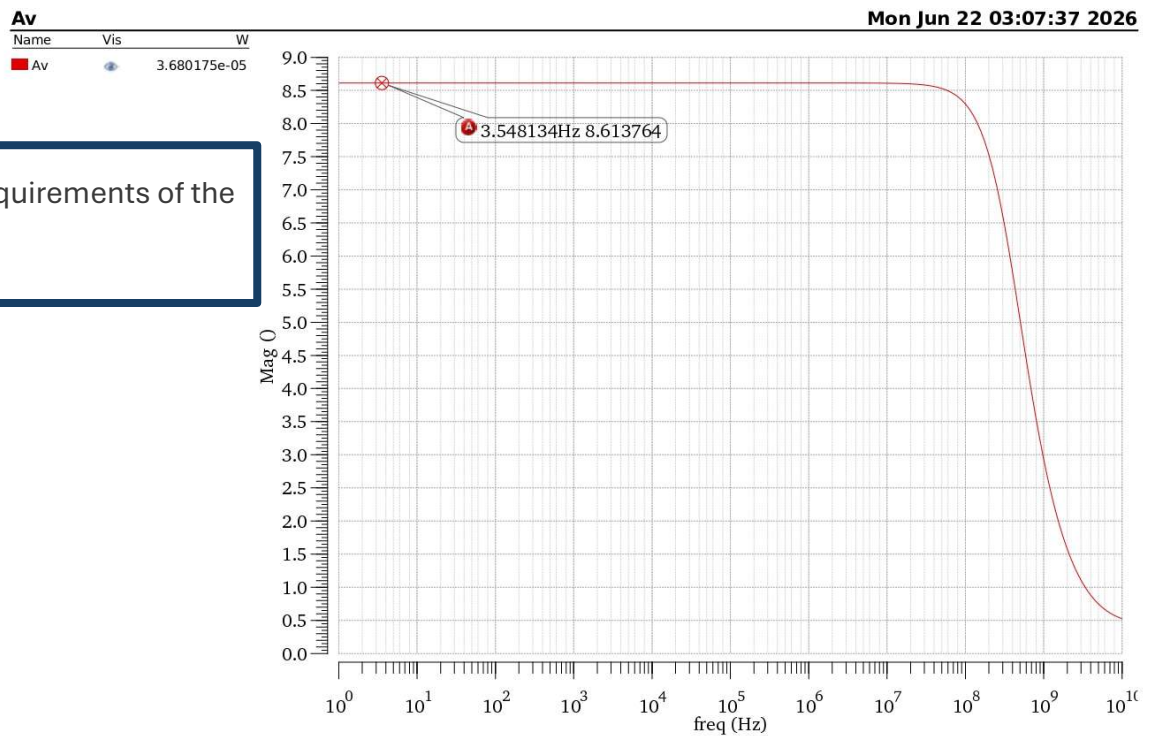
$$A_v = -g_m (R_D \parallel r_o) \quad R_D = 10.714 \text{ k}\Omega \quad r_o = 215.5 \text{ k}\Omega$$

$$R_D \parallel r_o = 10.207 \text{ k}\Omega \quad A_v = -843.915 \mu\text{S} \cdot 10.207 \text{ k}\Omega = -8.613$$

$$A_v \ll A_{v \text{ intrinsic}}$$

6) AC Analysis

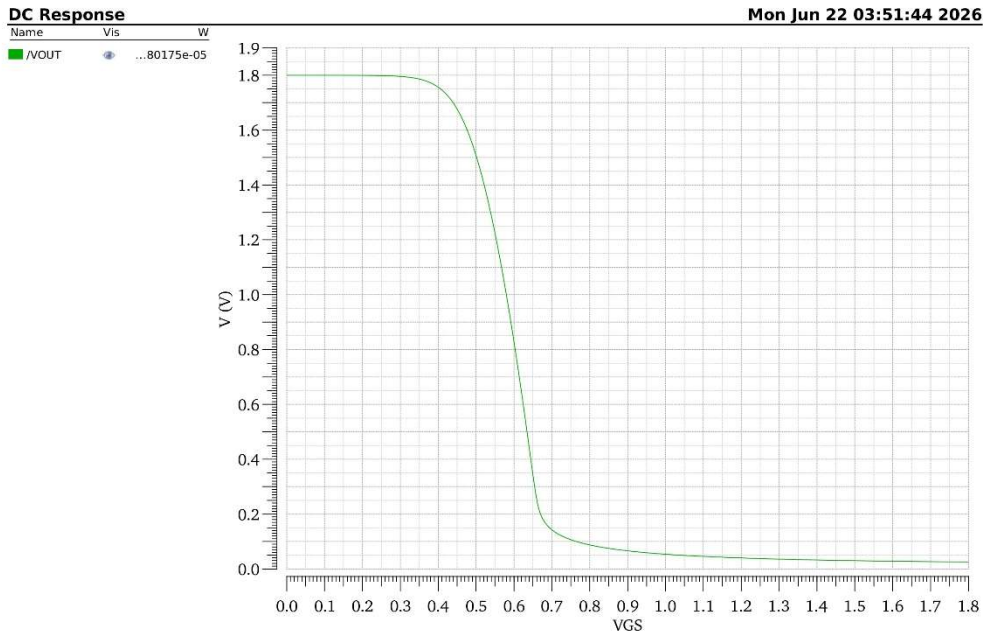
DC gain meets the requirements of the gain = 8.613



Gain Non-Linearity

1) DC Analysis

Transfer function



Choosing Analyses -- ADE L (1)

Analysis: ☐ tran ☒ dc ☐ ac ☐ noise
☐ xf ☐ sens ☐ dcmatch ☐ acmatch
☐ stb ☐ pz ☐ sp ☐ envlp
☐ pss ☐ pac ☐ pstb ☐ pnoise
☐ pxf ☐ psp ☐ qpss ☐ qpac
☐ qpnoise ☐ qpxf ☐ qpss ☐ hb
☐ hbac ☐ hbnoise ☐ hbasp

DC Analysis

Save DC Operating Point ☒
Hysteresis Sweep ☐

Sweep Variable

☐ Temperature
☒ Design Variable Variable Name:
☐ Component Parameter Select Design Variable
☐ Model Parameter

Sweep Range

☒ Start-Stop Start: Stop:
☐ Center-Span

Sweep Type

☒ Linear Step Size:
☐ Number of Steps

Add Specific Points ☐

Enabled ☐

Options...

2) Linearity of the relation

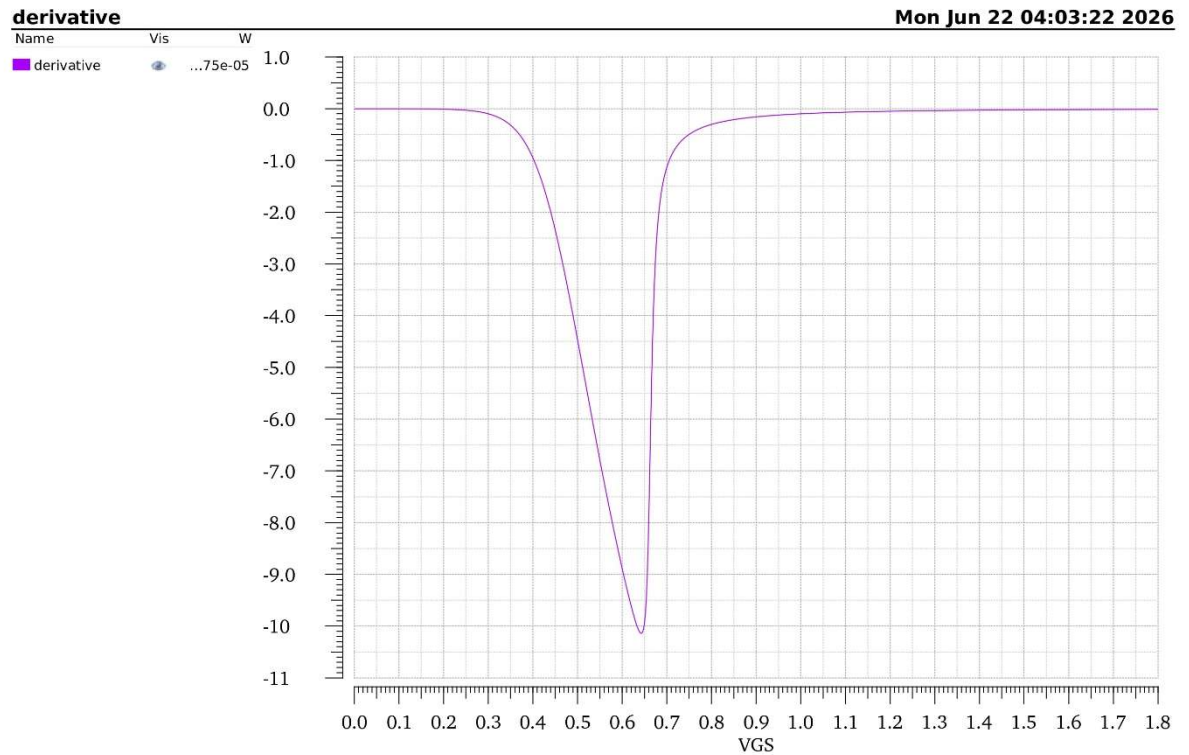
This relation is not linear

As there are three region first region is cutoff in which the output is V_{DD} this region is in which $I_D = 0$.

The second region is the saturation region in which the increase in V_{IN} decreases V_{OUT} as $V_{OUT} = V_{DD} - I_D \cdot R_D$ so as V_{IN} increases I_D increases and thus V_{OUT} decreases.

The third and last region is the triode region in which the mosfet gets out of saturation as the I_D is very large so V_{DS} becomes less than V_{GS} so it go into triode which acts as a voltage controlled resistance.

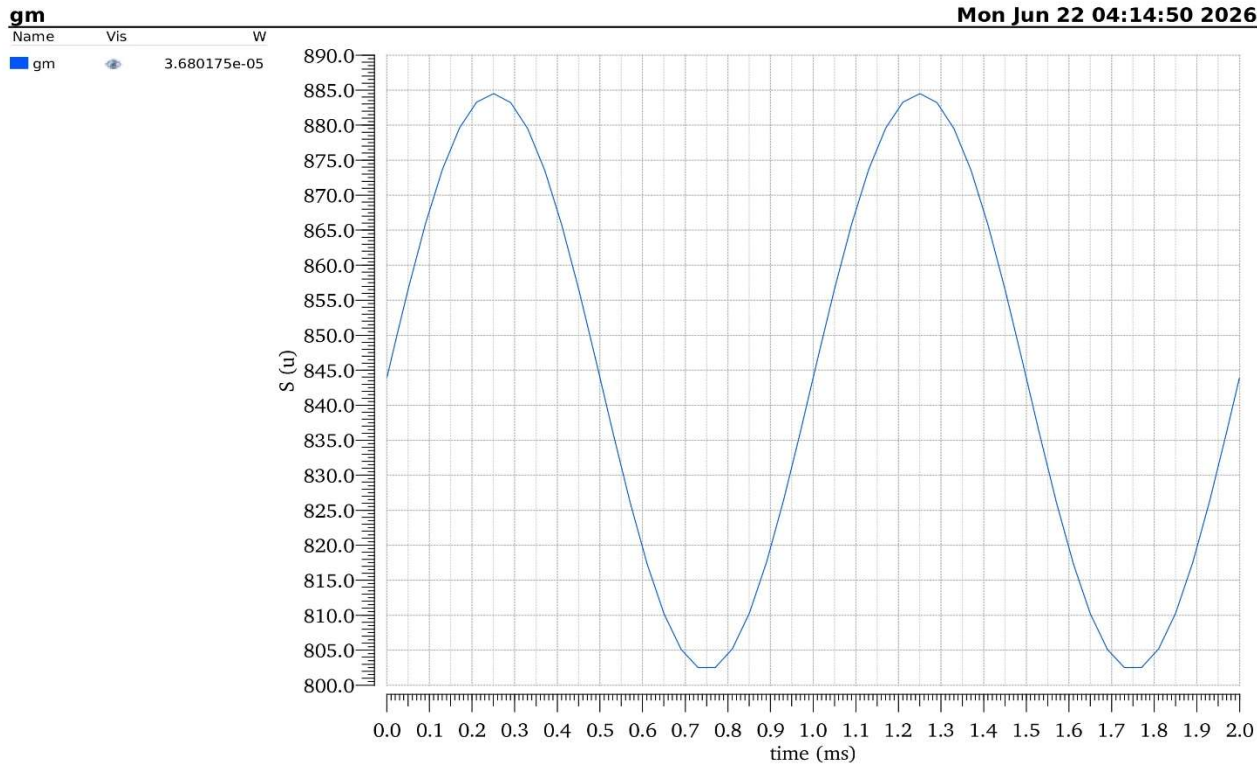
3) Derivative of VOUT



No the gain is not independent of the V_{IN} as V_{IN} determines the operating point which determines the gain as when V_{IN} changes the small signal parameters like g_m , r_o so it's not independent of V_{IN} , however we can do an approximation of the small signal model and linearize the gain around the operating point

4) Transient Analysis

5) gm vs time plot



comment

Yes, g_m varies with input signal and this means that g_m is dependent on the input signal.

6) Linearity of the Amplifier

A linear amplifier has a linear transfer function, but in our case the transfer function depends on the VGS (V_{IN}), so the gain is not linear and the amplifier is also not linear.

Edit Object Properties

Apply To: ☐ only current ☒ instance

Show: ☐ system ☒ user ☒ CDF

Browse Reset Instance Labels Display

Property	Value	Display
Library Name	analogLib	off
Cell Name	vsource	off
View Name	symbol	off
Instance Name	V2	off

Add Delete Modify

User Property	Master Value	Local Value	Display
Ignore	TRUE	S	off

CDF Parameter

Parameter	Value	Display
DC voltage	VGS V	off
Source type	sine	off
Frequency name 1		off
Frequency 1	1K Hz	off
Amplitude 1 (Vpk)	10m V	off
Phase for Sinusoid 1		off
Sine DC level		off
Delay time		off

Voltage source settings